

Question Number	Answer	Additional Guidance	Mark
4a	A description that includes two of the following: - find a suitable {mass / concentration / number of cells} of yeast (that will produce carbon dioxide) (1) - find a suitable {method for measuring carbon dioxide / method to measure oxygen consumption / redox indicator} (1) - find a suitable range of temperatures (1)	Accept suitable {concentration / mass} of sugar Accept suitable concentration / mass of mineral ions eg TTC / DCPIP / methylene blue Accept find a suitable timescale to measure the {volume of gas produced / oxygen consumption}	Exp 2

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4b	An answer that includes eight of the following: • clear statement of the dependent variable (1) • some description of apparatus used (1) • control of mass of yeast (1) • incubate for a set period of time and record {volume of carbon dioxide produced / movement of ink drop} (1) • five stated temperatures in a range of 5-55 °C (1) • two variables that need to be controlled (1) • description of how one of these variables is controlled (1) • repeats for each temperature or repeat the whole experiment (1) • method of calculating rate of respiration (1)	e.g. volume of carbon dioxide produced per unit time / volume of oxygen used per unit time / time for (named) redox indicator to change colour eg method of collecting (carbon dioxide) gas / respirometer with soda lime / tubes in a waterbath before mixing Accept {volume / concentration} of yeast (suspension) / number of yeast cells record time for {colour change of redox indicator / standard volume of (carbon dioxide) gas to be collected / ink drop to move standard distance} pH – buffer; {concentration / volume} of redox indicator; {mass / concentration} of glucose eg 10g sugar/ use of balance; {type / strain / species / age} of yeast; 1 divided by time taken for colour change / distance divided by time / volume divided by time	Exp 8

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4c	An answer that includes the following: • table for raw data with headings and units, and means calculated from repeats (1) • line graph format with labelled axes (1) • use of an appropriate correlation statistical test (1)		Exp 3

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4d	An answer that includes two of the following: • difficult to measure (small) values of the dependent variable (1) • difficult to prevent contamination of yeast cultures / hard to maintain aseptic conditions (1) • difficulties related to experimental design (1)	Accept difficult to recognise end point Accept uneven distribution of yeast cells at start of investigation / unequal numbers in each tube Accept yeast may change from aerobic to anaerobic respiration during investigation; build-up of waste products may {affect enzymes / slow rate of respiration}; at higher temperatures, gases (CO₂ and oxygen) expand, so this would affect the volume recorded (esp in respirometer); carbon dioxide is water soluble, so the volume of gas recorded may not be accurate	Exp 2

(Total for question 4 = 15 marks)